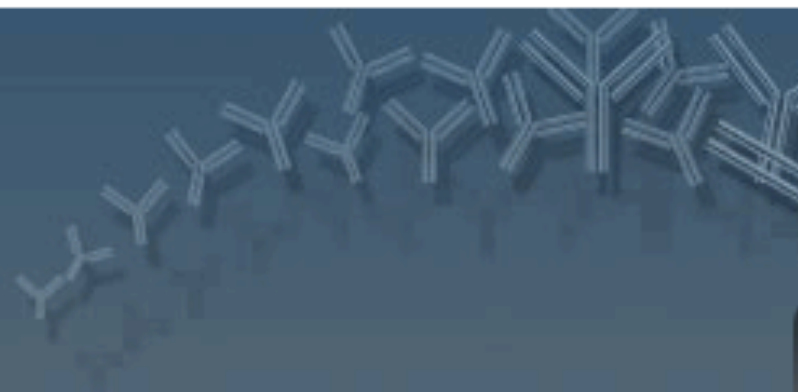






NIH HUMAN  
MICROBIOME  
PROJECT



[nature.com](#) ▶ [journal home](#) ▶ [archive](#) ▶ [issue](#) ▶ [article](#) ▶ [abstract](#)

---

## ARTICLE PREVIEW

---

[view full access options](#) ▶

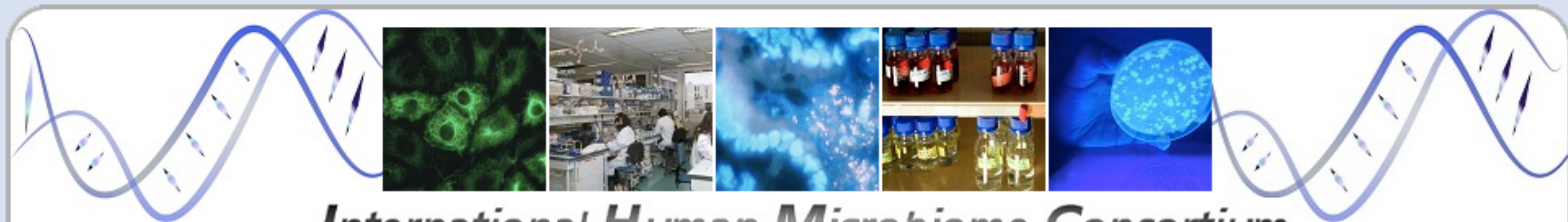
NATURE IMMUNOLOGY | ARTICLE



# Lymphotoxin regulates commensal responses to enable diet-induced obesity

Vaibhav Upadhyay, Valeriy Poroyko, Tae-jin Kim, Suzanne Devkota, Sherry Fu, Donald Liu, Alexei V Tumanov, Ekaterina P Koroleva, Liufu Deng, Cathryn Nagler, Eugene B Chang, Hong Tang & Yang-Xin Fu

[Affiliations](#) | [Contributions](#) | [Corresponding author](#)



*International Human Microbiome Consortium*

## ARTICLE PREVIEW

[view full access options](#) >

NATURE | LETTER



# Diet rapidly and reproducibly alters the human gut microbiome

Lawrence A. David, Corinne F. Maurice, Rachel N. Carmody, David B. Gootenberg, Julie E. Button, Benjamin E. Wolfe, Alisha V. Ling, A. Sloan Devlin, Yug Varma, Michael A. Fischbach, Sudha B. Biddinger, Rachel J. Dutton & Peter J. Turnbaugh

[Affiliations](#) | [Contributions](#) | [Corresponding author](#)

Nature (2013) | doi:10.1038/nature12820

Received 18 April 2013 | Accepted 29 October 2013 | Published online 11 December 2013

learn.genetics.utah.edu/content/microbiome/diseases/



🏠 > Current Issue > vol. 109 no. 2 > Sharon Greenblum, 594–599, doi: 10.1073/pnas.1116053109



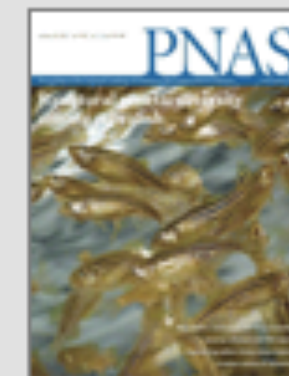
Performing your original search "microbiome disease links science" in *PNAS* retrieves [80 additional results](#).

# Metagenomic systems biology of the human gut microbiome reveals topological shifts associated with obesity and inflammatory bowel disease

Sharon Greenblum<sup>a</sup>, Peter J. Turnbaugh<sup>b</sup>, and Elhanan Borenstein<sup>a,c,d,1</sup>

Author Affiliations

## This Issue



January 10, 2012  
vol. 109 no. 2  
[Masthead \(PDF\)](#)  
[Table of Contents](#)

◀ PREV ARTICLE

NEXT ARTICLE ▶

Don't Miss

- Acne
- Antibiotic-associated diarrhea
- Asthma/allergies
- Autism
- Autoimmune diseases
- Cancer
- Dental cavities
- Depression and anxiety
- Diabetes
- Eczema
- Gastric ulcers
- Hardening of the arteries
- Inflammatory bowel diseases
- Malnutrition
- Obesity



Lymphotoxin receptor  
enable diet-induced

Vaibhav Upadhyay, Valeri

Alexei V Tumanov, Ekaterina

Hong Tang & Yang-Xin Fu

Affiliations | Contributions

Current Issue > vol. 109 no. 2 > Sharon Greenblum, 594–599, doi: 10.1073/pnas.1116053109



Performing your original search "microbiome disease links science" in PNAS retrieves 80 additional results.

## Metagenomic systems biology of the human gut microbiome reveals topological shifts associated with obesity and inflammatory bowel disease

Sharon Greenblum<sup>a</sup>, Peter J. Turnbaugh<sup>b</sup>, and Elhanan Borenstein<sup>a,c,d,1</sup>

Author Affiliations

### ARTICLE PREVIEW

[view full access options](#)

NATURE | LETTER



## Diet rapidly and reproducibly alters the human gut microbiome

Lawrence A. David, Corinne F. Maurice, Rachel N. Carmody, David B. Gootenberg, Julie E. Button, Benjamin E. Wolfe, Alisha V. Ling, A. Sloan Devlin, Yug Varma, Michael A. Fischbach, Sudha B. Biddinger, Rachel J. Dutton & Peter J. Turnbaugh

Affiliations | Contributions | Corresponding author

Nature (2013) | doi:10.1038/nature12820

Received 18 April 2013 | Accepted 29 October 2013 | Published online 11 December 2013

[learn.genetics.utah.edu/content/microbiome/disease/](http://learn.genetics.utah.edu/content/microbiome/disease/)



Human Microbiome Consortium

- Acne
- Antibiotic-associated diarrhea
- Asthma/allergies
- Autism
- Autoimmune diseases
- Cancer
- Dental cavities
- Depression and anxiety
- Diabetes
- Eczema
- Gastric ulcers
- Hardening of the arteries
- Inflammatory bowel diseases
- Malnutrition
- Obesity



# Bacterial cells can help tumors

## REPORT

## Potential role of intratumor bacteria in mediating tumor resistance to the chemotherapeutic drug gemcitabine

Leore T. Geller<sup>1,\*</sup>, Michal Barzily-Rokni<sup>2,\*</sup>, Tal Danino<sup>3,†</sup>, Oliver H. Jonas<sup>4,5</sup>, Noam Shental<sup>6</sup>, Deborah Nejman<sup>1</sup>, Nancy Gavert<sup>1</sup>, Yaara Zwang<sup>1</sup>, Zachary A. Cooper<sup>7,8,‡</sup>, Kevin Shee<sup>2</sup>, Christoph A. Thaiss<sup>9</sup>, Alexandre Reuben<sup>8</sup>, Jonathan Livny<sup>2</sup>, Roi Avraham<sup>10</sup>, Dennie T. Frederick<sup>11</sup>, Matteo Ligorio<sup>12</sup>, Kelly Chatman<sup>13</sup>, Stephen E. Johnston<sup>2</sup>, Carrie M. Mosher<sup>2</sup>, Alexander Brandis<sup>14</sup>, Garold Fuks<sup>15</sup>, Candice Gurbatri<sup>16</sup>, Vancheswaran Gopalakrishnan<sup>8</sup>, Michael Kim<sup>8</sup>, Mark W. Hurd<sup>17</sup>, Matthew Katz<sup>8</sup>, Jason Fleming<sup>8</sup>, Anirban Maitra<sup>18</sup>, David A. Smith<sup>2</sup>, Matt Skalak<sup>3</sup>, Jeffrey Bu<sup>3</sup>, Monia Michaud<sup>19</sup>, Sunia A. Trauger<sup>13</sup>, Iris Barshack<sup>20,21</sup>, Talia Golan<sup>21,22</sup>, Judith Sandbank<sup>21</sup>, Keith T. Flaherty<sup>12</sup>, Anna Mandinova<sup>2,23</sup>, Wendy S. Garrett<sup>2,19,24</sup>, Sarah P. Thayer<sup>25</sup>, Cristina R. Ferrone<sup>26</sup>, Curtis Huttenhower<sup>2,27</sup>, Sangeeta N. Bhatia<sup>2,28,29,30,31,32,33</sup>, Dirk Gevers<sup>2,§</sup>, Jennifer A. Wargo<sup>7,8</sup>, Todd R. Golub<sup>34,35,36,¶</sup>, Ravid Straussman<sup>1,¶,¶</sup>

“Certain bacteria express enzymes capable of metabolizing the cancer chemotherapeutic drug *gemcitabine* into an inactive form...an effect that was reversed by antibiotic treatment in mice. A high percentage of human pancreatic ductal adenocarcinomas contain the culprit bacteria (gammaproteobacteria). ...Efficacy of an existing therapy for this lethal cancer might be improved by co-treatment with antibiotics.”